Biospectra : ISSN: 0973-7057, Vol. 20(2), September, 2025, pp. 29-36

An International Biannual Refereed Journal of Life Sciences

Animal Science

ISSN : 0973-7057

Vermicomposting: A sustainable approach for waste management and soil health improvement

Pankaj Saini* & Sushma Jain

Department of Zoology, Vidya Bhawan Rural Institute, Udaipur, Rajasthan, India

Received : 24th April, 2025 ; Revised : 23rd May, 2025

DOI:-https://doi.org/10.5281/zenodo.18168397

Abstract- In this advanced era, people are developing and using advanced technologies day by day to fulfill their requirements, to get more products, and to be better than each other. But amidst all this, they have forgotten to protect the environment and are continuously making it worse by spreading waste materials as much as they can. In 2025, the World Bank and the Indian Ministry of Environment, Forests, and Climate Change (MoEFCC) emphasized the growing amount of trash generated, the lack of proper collection and treatment, and the urgent need for better waste management systems. The increasing amount of waste, like agricultural wastes, wetland waste, domestic wastes, municipal waste, animal wastes, and industrial wastes, day by day, has become a matter of the biggest concern. Vermicomposting is a useful and cost-effective way to overcome this problem by managing the wastes and converting them into a beneficial product. In this technique, worms degrade the wastes and enhance the nutrients in the end product. This end product makes soil better and enhances the growth of plants and nutrients in the yield. In this way, vermicomposting helps in biodegradable waste management, and other hand, it improves nutrition-rich, valuable crop production.

Keywords: Technologies, waste materials, environment, vermicomposting, manure

INTRODUCTION

One of the most promising waste recycling technologies is 'Vermicomposting', which is helpful in bioremediation of contaminated material and is considered a futuristic approach among the many waste recycling solutions available. Compost, which is prepared with a certain type of species of earthworm and used to enhance the process of waste conversion and to get a better product, is called 'Vermicompost', and this biotechnological process of making compost is called as 'Vermicomposting'. This technique is very fruitful in waste management and is easy to process at a very low cost. The vermicomposting process can be used in the transformation of waste into nutrient-rich manure by degrading waste material,[1] and it can also increase the microbial population in the end product as compared to the initial stage of it.

At the present time in agricultural fields, People are using chemicals in their farms to get more products. Chemicals can increase the food quantity but decrease the nutrition in food, and also decrease the soil fertility. Due to regular treatment of chemicals given to soil, soil becomes addicted to chemicals, and then more amount of chemicals are required to maintain the fertility of the soil and the productivity of food at the same level. Overuse of these chemicals affects beneficial soil microbes and is also responsible for water pollution, nutritive imbalance, contamination of ground and surface water. But application

*Corresponding author :
Phone : 7300217600
E-mail : pankajsaini42282@gmail.com
hodzoologyvbri@vidyabhawan.in

of vermicompost which is prepared with different type waste materials, to soil as soil amendment, it helps to improve soil fertility, neutralize soil pH, increase soil organic matter, promotes the microbial activity, alter porosity and structure of soil, increase water holding capacity, to help carbon sequestration, decrease soil erosion, inhibit the growth of plant pathogen, increase crop production, encourage organic farming, reduce environmental impact and also helps in solid waste management.[2] The end product of vermicomposting has essential micro and macronutrients such as P, N, K, Ca, Cu, Mg, Fe, Mn, and Zn.[3] Vermicompost is an organic fertilizer, a popular soil additive that has proven to efficient and cost-effective substitute for expensive synthetic pesticides and chemical fertilizers. Raza *et al.* (2024)[4] reviewed and evaluated that preparing for vermicomposting, N_2O (Nitrous oxide) emission did not increase as compared to conventional compost, and vermicompost is good for soil health and plant growth. Rameshwar and Argaw (2016)[5] also studied vermicomposting and found that this is an effective technology to utilize solid wastes.

Organic wastes can be rich in organic carbon and nutrients, and they can increase crop production and overall health of crops and soil by using them in vermicomposting rather than throwing them directly into the soil. Biodegradable garbage like agricultural waste, kitchen waste, wetland waste, and animal waste can pollute the environment by being thrown directly on the earth and can be harmful for humans and other animals.[6] Vermicompost can be used as a soil conditioner and enhances microbial populations, and supports sustainable agriculture. This is the best way for waste management and increases agricultural productivity.[7] So, vermi composting is a useful technique for repurposing domestic, commercial, municipal, and agricultural wastes, and it accelerates humification, improves the chemical and physical characteristics of material, and enhances the microbial population in soil.

Vermicomposting as a Waste Management Strategy

Agricultural Waste

Locally available materials like grass, teff, hair-cot bean, maize, and sorghum straw used in vermicomposting, increase the macro and micro nutrients of the plant. Grass-vermicompost has the highest level of nitrogen and phosphorus content and the highest potassium and calcium observed in teff, the highest magnesium observed in hair-cot bean, the highest iron observed in teff straw, the highest manganese observed in hair-cot bean, highest copper in sorghum.[8] Physical parameters like pH, conductivity, total organic carbon, etc., changed in vermicompost prepared with organic waste like rice straw and paper waste. Vermicompost prepared with substrates like maize stalk, wheat straw, sorghum stalk, soybean straw, hair-cot bean straw, faba bean straw, and finger millet straw with the help of earthworm *Eisenia foetida* and cow dung affects the physical and chemical properties of compost differently.[9] Paddy straw and rain tree litter wastes can also be used to influence the value of vermicompost. These wastes have different levels of nutrient content, like nitrogen, potassium, phosphorus, copper, zinc, manganese, and iron. Vermicompost made by only rain tree litter has a higher amount of nutrients.C: N ratio ranged between 9.71- 16.21. Vermicompost prepared with 100% rain tree litter has higher total nitrogen (2.31%), Potassium (0.863%), Phosphorus (1.185%), manganese (231.50 mg /kg), Iron (1072 mg/kg), copper (54.25 mg/kg), and Zinc (254 mg/kg).[10]

Combined treatment of biochar and vermicompost can increase the plant growth and leaf nutrients by improving the chemical properties like Total soluble solids, Total phenol content, beta-carotenoids, Titratable acidity, Ascorbic acid, and physical properties like fresh weight, diameter, length, and thickness of fruit.[11]

Vermicompost prepared with leonardite, perlite, cocopeat, and peat moss can improve plant growth. It enhances the concentration of nutrients in fruit and leaf.[12] Kureljušić *et al.* (2024)[13] studied that vermicompost prepared with apple pomace and wheat straw can decrease the heavy metal like Cu, Zn, Cd, and Pb.

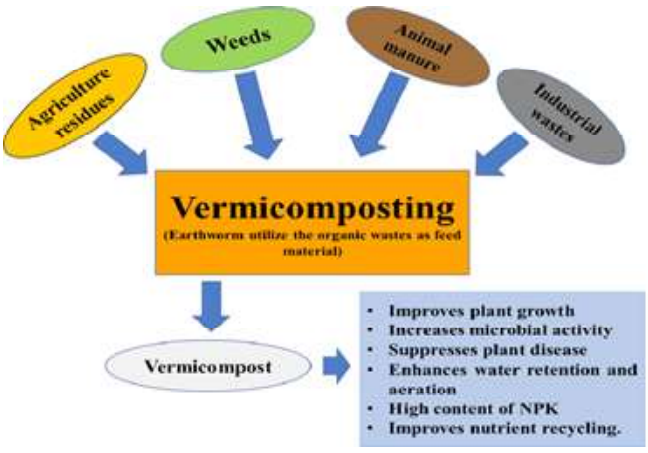


Fig. 1- Vermicomposting with different types of waste

Wetland Waste

Wetland waste can be harmful to living organisms. The *Eichhornia crassipes* (water hyacinth) plant is harmful to water living organisms by causing hypoxia. Vermicompost prepared with wetland plant wastes affects the plant growth and biomass. Raza, *et al*. (2024)[6] found that *Hydrocotyl vulgaris* and *Acorus calamus* wetland plants' combined vermicompost increases the soil organic matter (192% increase) and total nitrogen (92% increase). Combined vermicompost of *Cyperus alternifolius* and *Hydrocotyl vulgaris* increases the shoot biomass. Application of vermicompost and vermicompost tea prepared with macrophyte biomass can also increase the plant height, number of leaves, fruit quantity, and total yield of the plant. It has the potential to control the Fusarium wilt disease.[14]

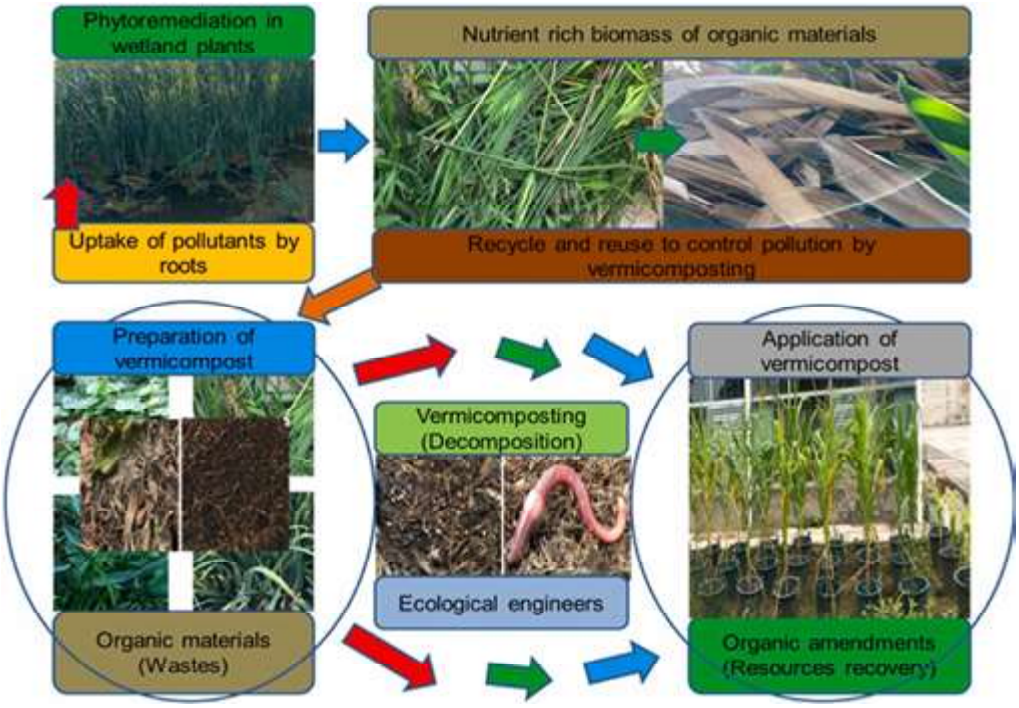


Fig. 2- Wetland Waste Treatment (Raza, Feyissa, *et al*., 2024)[6]

Animal Waste

Vermicomposting with *Eudrilus eugeniae* and got dung enriched by *Azolla pinnata* enhances the nitrogen, total carbon and potassium in prepared vermin- compost as compare to simple got dung vermicompost.[15] Shahiful Hizam *et al*., (2024)[16] reported that animal waste like chicken feathers can influence the nutrient content. Vermicomposting with mashroom media, banana trunk, chicken dung chicken feather and composting agent-earthworm and bacteria improved the nutrient quality of vermicompost.

Vermicompost of hop (*Humulus lupulus*) waste and horse manure influence the cocoon production and growth rate of *Eisenia andrei*. Hop waste and horse manure influence the number of earthworm, total biomass, numbers of cocoon, number of clitellated earthworm, rate of mortality and number of hatchling.[17]

Kitchen Waste

Vermicompost prepared with fine chopped household waste such as beet peel, carrot, fruit squash, banana, pumpkin, remains of tea and coffee are enriched in humic acid, phosphorous and nitrogen content. This vermicompost enhances the soil fertility, plant growth rate, energy of seed germination and plant height.[18] Walia and Kaur, (2024)[19] also studied that household and farm wastes vermicompost increase the soil quality, improve soil structure and used as soil conditioner.

Vermicompost quality can be increased by using vegetable and tea leaf as substrates with cow dung in different ratio, which can influence the organic carbon, NPK level, C: N ratio, moisture content, pH and electrical conductivity.[20]

Mohammad Alipour *et al*., (2023)[21] evaluated that usage of potato peel, carrot pulp, saw dust and vegetables

used in vermicompost can furnish most favorable conditions to *Eisenia foetida* for reproduction and growth and very qualitative vermicompost can be prepared with higher micronutrients. Poultry and vegetable wastes showed significant increase in nitrogen, phosphorus and potassium content of vermicompost. Vermicompost is pollutant free product.[22]



Fig. 3 - Different Types of Kitchen Wastes

Municipal Solid Waste (MSW)

Municipal waste negatively affects human health and environment. Many people used to fired the MSW for reduce the waste but it is more dangerous for environment and humans due to releasing hazardous gases from burning wastes.[23] Chetankumar *et al*., (2020)[24] reported that vermicompost prepared with MSW improve the growth of chickpea plant by improving plant height and pod formation. It also increases the protein content of seed and yield of protein. *Eisenia foetida* degrade the municipal biodegradable solid waste and transform it into nutrient rich, environment friendly biofertilizer and helps in overcome the hazardous of waste.[25]

Industrial Waste

India gets 70% of energy from thermal power plants, which produce the high amount of fly ash as by-product. Fly ash vermicompost enhances the seed germination of tomato and brinjal. Process of vermicomposting manage the CCR (coal combustion residues), which is by-product of burning coal in fly ash.[26] Vermicomposting process can be used in remediating tailing soil of coal mine, which enhances the soil fertility, plant growth and decrease the risk of heavy metal contamination. It can help to mitigate the hazardous environmental impacts of coil mining.[27] It can also reduce the textile waste and convert in to nutrient-rich compost.[28]

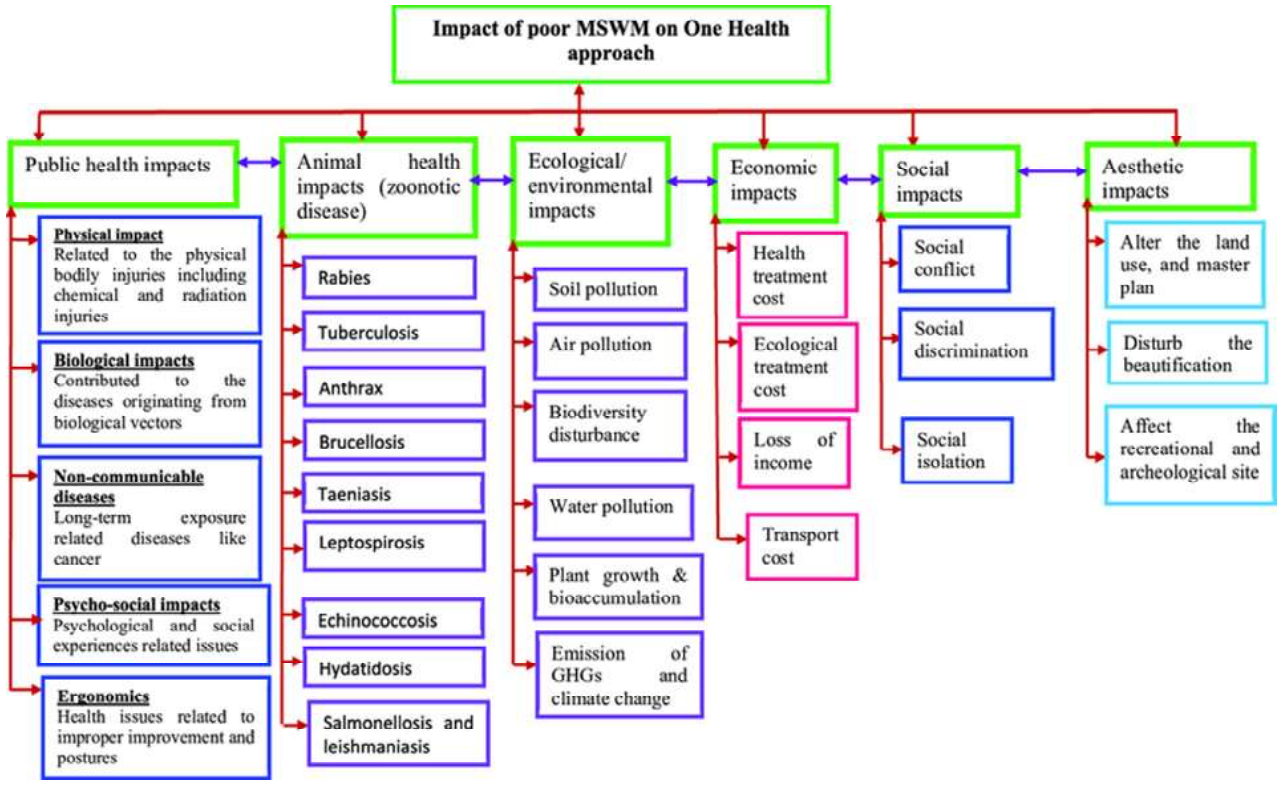


Fig. 4 - Municipal solid wastes' effect on human health and environment (Gebrekidan *et al*., 2024)[29]

Benefits of Vermicompost

Plant Growth and Crop Yield

Vermicomposting influences the plant's overall physical and chemical growth and helps to get more beneficial results.[30] It also increases the survival percentage of plant[31] and increases the nutrient uptake by plant[32].

It increases the fresh weight of leaves, length of leaf, number of leaves per plant, dry weight of leaves, area of leaf, spike length, fresh weight of spike, dry weight of spike, rachis length, flower diameter, number of florets, petiole length, fruit's ripening period, fresh weight, diameter, length and thickness of fruit, dry and fresh weight of root and shoot, nutrient content like N, P, K, Zn, Fe, Cu, Mn, chlorophyll-a, chlorophyll-b, carotenoids, carbohydrates, poly-phenols, SOD (superoxide dismutase) anti-oxidant enzyme. Vermicompost also affects the photosynthetic efficiency of plants.

Application of vermicompost in seed cultivation, increases the nutritional value of microgreens by balancing glucosinolates and phenolics.[33] It also affects the water regulation and physiological activities by enhancing stomatal conductance.[34] Phytopathogen resistance seed-beds can be produced by repressing the growth of phytopathogen and it can enhances the healthy and disease-resistant plant growth.[35]

Soil Fertility and Structure

Good soil health and fertility are essential for agricultural system performance because it directly affect the ecosystem sustainability and plant health. Vermicompost improves the soil health[36] and improve the soil quality by enhancing nutrient level in soil[37]. Vermicompost increase the saprophytic microbes multiplication in soil, which can enhance the control on phyto-pathogens.[38] It improves the nutrients like N, P, K, organic matter, enzymatic activity of Urease, catalase, sucrase and alkaline phosphatase, microbial activity, pore size, water holding capacity, soil fertility, soil structure, cation exchange capacity of soil.

Yuan *et al*., (2025)[39] evaluate that vermicompost employed soil enhances the carbon cycling by enhancing soil carbon sequestration with Phage- encoded AMG (auxiliary metabolic genes).

Environmental Benefits

Vermicomposting technique reduces the hazardous effect of solid wastes and decrease the soil pollution. It is an effective method for solid waste management. Vermicomposting technique reduces the greenhouse gas emission which helps in elevating soil health.[40] Specially CH_4 and N_2O greenhouse gas emission can be reduced with vermicomposting.[41]

Heavy Metal Remediation

Aransiola *et al*., (2024)[42] found that vermitechnology is a useful technique to decrease the contamination of heavy metal from soil. Vermicompost application into soil can decrease the uptake of Cadmium by plants from the contaminated soil. It enhances the fungal community such as *Chlorophyta, Ascomycota, Basidiomycota, Ciliophora, and Glomeromycta* of soil and improves the sustainability of crop productivity.[43] Vermicomposting technique also reduce the Arsenic[44] and copper toxicity[45] from toxic mine wastes. Wu *et al.*, (2025)[46] also studied that using of vermicompost prepared with different substrates promoted the *Apium graviolens* plant growth and increases the biomass and limiting the Cd migration from soil to plant.

Pesticide Remediation

Vermicomposting technique reduces the pesticides effect by degrading and adsorbing pesticide with bio-mixtures. It is a better solution for waste water treatment.[47] It removes the pesticides like Acetochlor (herbicide) and Endosulfan Lactone from contaminated soil and overcome the bad impact of pesticide on environment.

Antibiotic Resistance Gene Reduction

Vermicomposting can be used in mitigating antibiotic resistance in organic waste treatment and the study of the impact of nano-biochar on phage-host interactions and antibiotic resistance genes proliferation.[48] This can reduce the environmental risks by reducing the antibiotic- resistant bacteria which are harmful for environment.[49]

CONCLUSION

Vermicomposting minimize the waste material hazardous and overcome the environmental pollution which spreading by different types wastes. It degrades the biodegradable solid waste and prepared the good quality manure. This is an eco-friendly and cost effective technique for waste management and balancing the nutrient content in soil and plants. Vermicomposting process enhances the physical and chemical properties of plants. It positively effects the crop production. This is the cost effective bio-technique to reduce the pollution from environment by waste management.

REFERENCES

1. **Lirikum, Kakati, L. N., Thyug, L., & Mozhui, L. 2022.** Vermicomposting: An eco-friendly approach for waste management and nutrient enhancement. *Tropical Ecology*, **63(3):** 325–337.
2. **Johnson Jeyaraj, N., & Sankararajan, V. 2025.** Sustainable utilization of solid waste byproducts in agricultural amendments as soil conditioner: A state-of-the-art review. *International Review of Applied Sciences and Engineering*.
3. **Ramnarain, Y. I., Ansari, A. A., & Ori, L. 2019.** Vermicomposting of different organic materials using the epigeic earthworm Eisenia foetida. *International Journal of Recycling of Organic Waste in Agriculture*, **8:** 23–36.
4. **Raza, S. T., Rong, L., Rene, E. R., Ali, Z., Iqbal, H., Sahito, Z. A., & Chen, Z. 2024.** Effects of vermicompost preparation and application on waste recycling, NH_3, and N_2O emissions: A systematic review on vermicomposting. *Environmental Technology & Innovation*, **35:** 103722.
5. **Rameshwar, H. Y., & Argaw, A. 2016.** Manurial value of khat waste vermicompost from Awday, Harar town, Ethiopia. *International Journal of Recycling of Organic Waste in Agriculture*, **5(2):** 105–111.
6. **Raza, S. T., Feyissa, A., Li, R., Rene, E. R., Ali, Z., Iqbal, H., Sahito, Z. A., & Chen, Z. 2024.** Emerging technology effects on combined agricultural and eco-vermicompost. *J. of Env.Management*, **352:** 120056.
7. **Almaramah, S. B., Abu-Elsaoud, A. M., Alteneiji, W. A., Albedwawi, S. T., El-Tarabily, K. A., & Al Raish, S. M. 2024.** The impact of food waste compost, vermicompost, and chemical fertilizers on the growth measurement of red radish (*Raphanus sativus*): A sustainability perspective in the United Arab Emirates. *Foods*, **13(11):** 1608.
8. **Geremu, T., Hailu, H., & Diriba, A. 2020.** Evaluation of Nutrient Content of Vermicompost Made from Different Substrates at Mechara Agricultural Research Center on Station, West Hararghe Zone, Oromia, Ethiopia. *Ecology and Evolutionary Biology*, **5(4):** 125.
9. **Dube, B., Chimdessa, D., & Sori, G. 2025.** Preparation and Characterization of Vermicompost Nutrients Made from Different Sources of Organic Materials. *Science Journal of Analytical Chemistry*, **13(1):** 15–23.
10. **Jadhav, A. B., Gosavi, A. B., Jadhav, S. A., & Patil, A. V. 2024.** Influence of Different Substrates on Nutrient Composition of Vermicompost and Vermiwash. *Int. Journal of Plant & Soil Science*, **36(6):** 340–352.
11. **EL-Mogy, M. M., Adly, M. A., Shahein, M. M., Hassan, H. A., Mahmoud, S. O., & Abdeldaym, E. A. 2024.** Integration of Biochar with Vermicompost and Compost Improves Agro-Physiological Properties and Nutritional Quality of Greenhouse Sweet Pepper. *Agronomy*, **14(11):** 2603.
12. **Erdal, Ý., & Aktaþ, H. 2025.** Comparison of the Perlite, Leonardite, Vermicompost and Peat Moss and Their Combinations with Cocopeat as Tomato Growing Media. *Journal of Soil Science and Plant Nutrition*.
13. **Kureljušiæ, J. M., Veskoviæ Moraèanin, S. M., Đukiæ, D. A., Mandiæ, L., Đuroviæ, V., Kureljušiæ, B. I., & Stojanova, M. T. 2024.** Comparative Study of Vermicomposting: Apple Pomace Alone and in Combination with Wheat Straw and Manure. *Agronomy*, **14(6):** 1189.
14. **Yatoo, A. M., Ali, M. N., Baba, Z. A., Alsohim, A. S., Muthukumaran, M., & Sayyed, R. Z. 2024.** Effect of macrophyte biomass-based vermicompost and vermicompost tea on plant growth, productivity, and biocontrol of *Fusarium* wilt disease in tomato. *Biocatalysis and Agr. Biotechnology*, **60:** 103320.
15. **Gayathry, G., Natarajan, K., Jayalakshmi, T., Kalaichelvi, K., Bharathikumar, K., Jayakumar, J., & Sugumaran, M. P. 2025.** Enhancing Vermicompost Quality with *Azolla pinnata*: Enriched Goat Dung for Sustainable Agriculture. *Journal of Advances in Microbiology*, **25(1):** 1–7.
16. **Shahiful Hizam, N., Omar, N., Sabri, N., Samsuddin, N., & Zainol Abidin, M. 2024.** Evaluation of Physicochemical Properties and Nutrient Composition of Chicken Feather Vermicompost for Sustainable Waste Management System. *IOP*

Conference Series: Earth and Environmental Science, **1426(1):** 012014.

17. **González-Moreno, M. Á., García Graciante paraluceta, B., Marcelino Sádaba, S., Prieto Cobo, E., & Seco, A. 2024.** A biological insight of hops wastes vermicomposting by Eisenia Andrei. *J. of M. Cycles and Waste Man.* **26(1):** 444–454.
18. **Kallas, E., Rodikova, A., Maron, T., Myasnikov, A., & Gulik, E. 2025.** Efficiency of vermicompost in a microplot field experiment. *BIO Web of Conferences*, **160:** 01021.
19. **Walia, S. S., & Kaur, T. 2024.** Different Sources of Vermicompost: Vermicomposting from Household Waste—Vermicomposting from Farm Waste. In S. S. Walia & T. Kaur, *Earthworms and Vermicomposting* (pp. 61–71). Springer Nature Singapore.
20. **Shrestha, G., Gwachha, S., & Shrestha, M. B. 2024.** Comparison of Vermicomposting Quality using Different Food beds. *J. of Env. Sciences*, **10:** 1–9.
21. **Mohammad Alipour, S., Fataei, E., Nasehi, F., & Imani, A. A. 2023.** Vermicompost quality and earthworm reproduction in different organic waste substrates. *International Journal of Recycling of Organic Waste in Agriculture*, **12(3).**
22. **Joseph, P. V. 2019.** Efficacy of different substrates on vermicompost production: A biochemical analysis. In *Organic Fertilizers-History, Production and Applications*. IntechOpen
23. **Walosik, A., Szajner, J., Pawlas, K., & eber-Dzikowska, I. 2021.** Municipal waste and related health risks. *J. of Elementology*, **26(3):** 573–582.
24. **Chetankumar, C., Vaidya, P. H., & Zade, S. P. 2020.** Effect of Vermicomposting and Composting of Municipal Solid Waste (MSW) on Growth, Yield and Quality of Chickpea. *International Journal of Environment and Climate Change*, 126–136.
25. **Mane, V. B., Kanase, S. S., Sawale, N. S., Bandsode, A. K., & Suryawanshi, M. A. 2024.** Application of earthworm Eisenia fetida in waste management in municipal solid waste. *Environment, Development and Sustainability*, 1–19.
26. **Usmani, Z., Kumar, V., Gupta, P., Gupta, G., Rani, R., & Chandra, A. 2019.** Enhanced soil fertility, plant growth promotion and microbial enzymatic activities of vermicomposted fly ash. *Sci. Reports*, **9(1):** 10455.
27. **Das, K., Dansena, S. K., Pottipati, S., Maturi, K. C., Dhal, G. C., & Kalamdhad, A. S. 2025.** Sustainable remediation of abandoned coal mines using vermicompost: A case study in Ledo coal mine, India. *Environmental Geochemistry and Health*, **47(2):** 51.
28. **Shanmugam, V., Subbaiyan, B., Shakshi, P., Begum, B., Selvam, R., Gnanendra, T., & Balasubramanian, R. 2025.** Vermicomposting; a potential technology, facilitating sustainable textile waste management. *J. of M. Cycles and Waste Man.* 1–21.
29. **Gebrekidan, T., Weldemariam, N., Degefa, H., Gebremedhin, G., & Weldemariam, A. 2024.** Impact of improper municipal solid waste management on fostering One Health approach in Ethiopia Challenges and opportunities: A systematic review. *Science in One Health*, **3:** 100081.
30. **Rashidy, R. I., Abdel-Rahman, S. S., Soliman, W. S., & Gahory, A. M. 2024.** Effect of vermicompost as a growth media on tuberose (*Polianthes tuberosa* L.) plant. *Aswan University Journal of Environmental Studies*, **5(4):**483–494.
31. **Anagha, P. K., Prakasha, D. P., Anand, N., Vijay, M., Ryavalad, S., & Sree, M. R. 2025.** Enhancing Dragon Fruit (*Hylocereus undatus*) Propagation Efficiency through Growing Media. *International Journal of Economic Plants*, **12(Jan, 1):** 01–05.
32. **Ramazanoglu, E. 2024.** Effects of vermicompost application on plant growth and soil enzyme activity in wheat (*Triticum aestivum* L.) monitored by thermal imaging. *Cogent Food & Agriculture*, **10(1):** 2373872.
33. **García-Pérez, P., De Gregorio, M. A., Capri, E., Zengin, G., & Lucini, L. 2025.** Unleashing the nutritional potential of Brassica microgreens: A case study on seed priming with Vermicompost. *Food Chemistry*, **475:** 143281.
34. **Oyege, I., & Balaji Bhaskar, M. S. 2025.** Sustainable Soybean Production Using Residual Vermicompost Inputs in Corn-Soybean Rotation. *Environments*, **12(9):**333.

35. **Alshehrei, F., Al-Enazi, N. M., & Ameen, F. 2025.** Vermicomposting amended with microalgal biomass and biochar produce phytopathogen-resistant seedbeds for vegetables. *Biomass Conversion and Biorefinery*, **15(2):** 1795–1802.

36. **Manzoor, A., Naveed, M. S., Ali, R. M. A., Naseer, M. A., Maqsood, U.-H., Saqib, M., Hussain, S., & Farooq, M. 2024.** Vermicompost: A potential organic fertilizer for sustainable vegetable cultivation. *Scientia Horticulturae*, **336:** 113443.

37. **Teshome Yitbarek, Rehima Redi, & Solomon Kibebew. 2025.** Enhancing Productivity of Bread Wheat (*Triticcum aestivum* L.) Through Integrating Vermicompost and N fertilizer in Meskan District, Central Ethiopia. *Jefore Ethiopian Journal of Applied Sciences*.

38. **Angmo, D., Singh, J., Rashid, F., Sharma, P., Thakur, B., Singh, S., & Vig, A. P. 2024.** Vermiremediation of organic wastes: Vermicompost as a powerful plant growth promoter. In *Earthworm Technology in Organic Waste Management* (pp. 59–77). Elsevier.

39. **Yuan, S., Wu, Y., Balcazar, J. L., Wang, D., Zhu, D., Ye, M., Sun, M., & Hu, F. 2025**. Expanding the potential soil carbon sink: Unraveling carbon sequestration accessory genes in vermicompost phages. *App. and Env. Micr.*, **91(4):** e00296-25.

40. **Kumar, A., Pandey, M., Srivastava, A., Pant, K., Gupta, A., & Ranjan, P. 2025.** Global Trends and Advances in Vermicomposting: A Bibliometric Analysis of its Multidisciplinary Applications. *Circular Economy and Sustainability*, 1–17.

41. **Li, J., Huang, Y., Wang, J., Zhang, Y., & Chen, Y. 2025.** Vermiculite changed greenhouse gases emission and microbial community succession in vermicomposting: Particle size investigation. *Bioresource Technology*, **416:** 131769.

42. **Aransiola, S. A., Josiah, I. U. J., Abioye, O. P., Bala, J. D., Rivadeneira-Mendoza, B. F., Prasad, R., Luque, R., Rodríguez-Díaz, J. M., & Maddela, N. R. 2024.** Micro and vermicompost assisted remediation of heavy metal contaminated soils using phyto extractors. *Case Studies in Chem. and Env. Eng*. **9:** 100755.

43. **Iqbal, A., Hussain, Q., Mo, Z., Hua, T., Mustafa, A. E.-Z. M. A., & Tang, X. 2024.** Vermicompost Supply Enhances Fragrant-Rice Yield by Improving Soil Fertility and Eukaryotic Microbial Community Composition under Environmental Stress Conditions. *Microorganisms*, **12(6):** 1252.

44. **Chakraborty, S., Banerjee, S., Kumar, S., Ghosh, S., Mukherjee, P., Das, S., & Bhattacharyya, P. 2025.** Biofortified vermicompost: Exploring bacterial community dynamics and enzymatic pathways through bacteriome analysis for arsenic bioremediation in mine waste. *Environmental Pollution*, **382:** 126795.

45. **Jabar, A., Athar, M., Mehmood, K., Ishfaq, T., Bashir, S., Iqbal, Z., Hussain, M., Zahra, A., Sherani, J., & Khan, S. 2025.** Evaluation of copper stress on maize and its remediation by application of compost and vermicompost. *Asian Journal of Agriculture and Biology*, **2025(01).**

46. **Wu, X., Xu, J., Ji, T., Huang, H., Hao, M., Ding, Y., & Zhu, W. 2025.** *Effects of Functional Vermicompost Application on Cd Migration Control and Environmental Impacts in a Soil-Celery System: A Field Study*.

47. **Clebot, A. C., Magni, F. V., Repetti, M. R., Zalazar, C. S., & Fernandez, M. E. 2025.** Adsorption, degradation, and synergistic effects of pesticides in biomixtures containing vermicompost from agricultural and urban waste. *Journal of Water Process Engineering*, **77:** 108384.

48. **Xie, T., Lin, D., Cai, X.-D., Ma, L.-J., Wang, L., Cai, T.-G., Ye, Y.-Q., Shen, L.-Q., Sun, M.-M., & Ye, M. 2025.** Nano-biochar regulates phage–host interactions, reducing antibiotic resistance genes in vermicomposting systems. *Proceedings of the National Academy of Sciences*, **122(34):** e251198 6122.

49. **Li, Z., Yang, F., Han, B., Zhao, R., Yang, M., & Zhang, K. 2025**. Vermicomposting significantly reduced antibiotic resistance genes in cow manure even under high tetracycline concentrations. *Bioresource Technology*, **419:** 132002.

***